Arrival · Supported

Respond to the command word.

Select a value or mechanism that supports the answer.

Improve a claim by adding an appropriate limitation.

Four checks. Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words. Choose one route; do not complete all three.

1. Describe

What does “describe” ask you to do?

Help: Read or point first. State the pattern or what happens.

2. Explain

What must an explanation add?

Help: Read or point first. A scientific reason or link.

3. Calculate

What should accompany a calculated answer?

Help: Read or point first. Working and any appropriate units.

4. Evaluate

What should an evaluation consider?

Help: Read or point first. Evidence and limitations, not just an opinion.

My response: _____

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Independent · Supported

Respond to the command word.

Select a value or mechanism that supports the answer.

Improve a claim by adding an appropriate limitation.

Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words. Choose one route; do not complete all three.

1. Explain

Improve the mitosis answer in two linked sentences.

Help: Use two linked sentences: DNA is copied during ___. The copies then ___ so each new nucleus ___.

2. Calculate

Show the working for 7/20 as a percentage.

Help: Write the fraction 7/20. Divide 7 by 20, then multiply by 100. Show the intermediate decimal.

3. Evaluate

Replace the safety claim with a qualified conclusion and a relevant improvement.

Help: Use: the study suggests ___, but does not establish ___. A useful improvement would be ___ because ___.

My response: _____

Use one route within the stated stage time. Keep your own original response.

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Exit · Supported

Respond to the command word.

Select a value or mechanism that supports the answer.

Improve a claim by adding an appropriate limitation.

1. Improve

Improve “the cells grew” using a supplied change from 2.0 to 2.8 mm.

Help: Use: my answer is ___ because ___. Point to a model or card first. Use a value and unit; do not invent a cause.

2. Explain

Why is “because it does” not a useful explanation of identical daughter nuclei?

Help: Use: my answer is ___ because ___. Point to a model or card first. Add copying and separation.

My response: _____

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Arrival · Independent

Respond to the command word.

Select a value or mechanism that supports the answer.

Improve a claim by adding an appropriate limitation.

Four checks. Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words. Choose one route; do not complete all three.

1. Describe

What does “describe” ask you to do?

Help: State the pattern or what happens.

2. Explain

What must an explanation add?

Help: A scientific reason or link.

3. Calculate

What should accompany a calculated answer?

Help: Working and any appropriate units.

4. Evaluate

What should an evaluation consider?

Help: Evidence and limitations, not just an opinion.

My response: _____

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Independent · Independent

Respond to the command word.

Select a value or mechanism that supports the answer.

Improve a claim by adding an appropriate limitation.

Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words. Choose one route; do not complete all three.

1. Explain

Improve the mitosis answer in two linked sentences.

Help: Use before / then / so.

2. Calculate

Show the working for 7/20 as a percentage.

Help: Divide, multiply by 100, add %.

3. Evaluate

Replace the safety claim with a qualified conclusion and a relevant improvement.

Help: Match the improvement to short follow-up.

My response: _____

Use one route within the stated stage time. Keep your own original response.

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Exit · Independent

Respond to the command word.

Select a value or mechanism that supports the answer.

Improve a claim by adding an appropriate limitation.

1. Improve

Improve “the cells grew” using a supplied change from 2.0 to 2.8 mm.

Help: Use a value and unit; do not invent a cause.

2. Explain

Why is “because it does” not a useful explanation of identical daughter nuclei?

Help: Add copying and separation.

My response: _____

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Arrival · Aspire

Respond to the command word.

Select a value or mechanism that supports the answer.

Improve a claim by adding an appropriate limitation.

Four checks. Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words. Choose one route; do not complete all three.

1. Describe

What does “describe” ask you to do?

Help: State the pattern or what happens.

2. Explain

What must an explanation add?

Help: A scientific reason or link.

3. Calculate

What should accompany a calculated answer?

Help: Working and any appropriate units.

4. Evaluate

What should an evaluation consider? Give a reason.

Help: Evidence and limitations, not just an opinion.

My response: _____

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Independent · Aspire

Respond to the command word.

Select a value or mechanism that supports the answer.

Improve a claim by adding an appropriate limitation.

Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words. Choose one route; do not complete all three.

1. Explain

Improve the mitosis answer in two linked sentences.

Help: Use before / then / so.

2. Calculate

Show the working for 7/20 as a percentage.

Help: Divide, multiply by 100, add %.

3. Choose evidence

A response lists four numbers but explains none. How would you improve it without making it much longer?

Help: Select the most relevant comparison and link it to the conclusion.

My response: _____

Use one route within the stated stage time. Keep your own original response.

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Exit · Aspire

Respond to the command word.

Select a value or mechanism that supports the answer.

Improve a claim by adding an appropriate limitation.

1. Improve

Improve “the cells grew” using a supplied change from 2.0 to 2.8 mm.

Help: Use a value and unit; do not invent a cause.

2. Choose evidence

A response lists four numbers but explains none. How would you improve it without making it much longer?

Help: Select the most relevant comparison and link it to the conclusion.

My response: _____

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Word page

Respond to the command word.

Select a value or mechanism that supports the answer.

Improve a claim by adding an appropriate limitation.

Word	Meaning
Describe	Say what happens or what the evidence shows.
Explain	Connect a statement to a scientific reason.
Calculate	Use values and show the working.
Evaluate	Weigh evidence and identify a limitation.
Evidence	A value, observation or source supporting a claim.
Independent event	An outcome that does not determine the next outcome.

Reference for teaching or later review. Use the command word to choose the kind of answer.

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Knowledge organiser

Respond to the command word.

Select a value or mechanism that supports the answer.

Improve a claim by adding an appropriate limitation.

Use command words, selected evidence and a clear scientific link to improve exam-style responses.

Copy DNA before mitosis. Each daughter cell retains the chromosome set.

Stem cell evidence needs benefits, risks and uncertainty.

A gene is a section of DNA. DNA has complementary base pairs: A with T, and C with G.

A genetic cross predicts a chance for each fertilisation.

Reference for teaching or later review. Choose evidence and add the scientific link.

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Starter

Respond to the command word.

Select a value or mechanism that supports the answer.

Improve a claim by adding an appropriate limitation.

Which answer best explains identical daughter nuclei?

A. They are identical because they match

B. DNA is copied, then chromosome copies are separated into two complete sets

C. The cell gets bigger

Keep your first choice. Give a reason before the answer is checked.

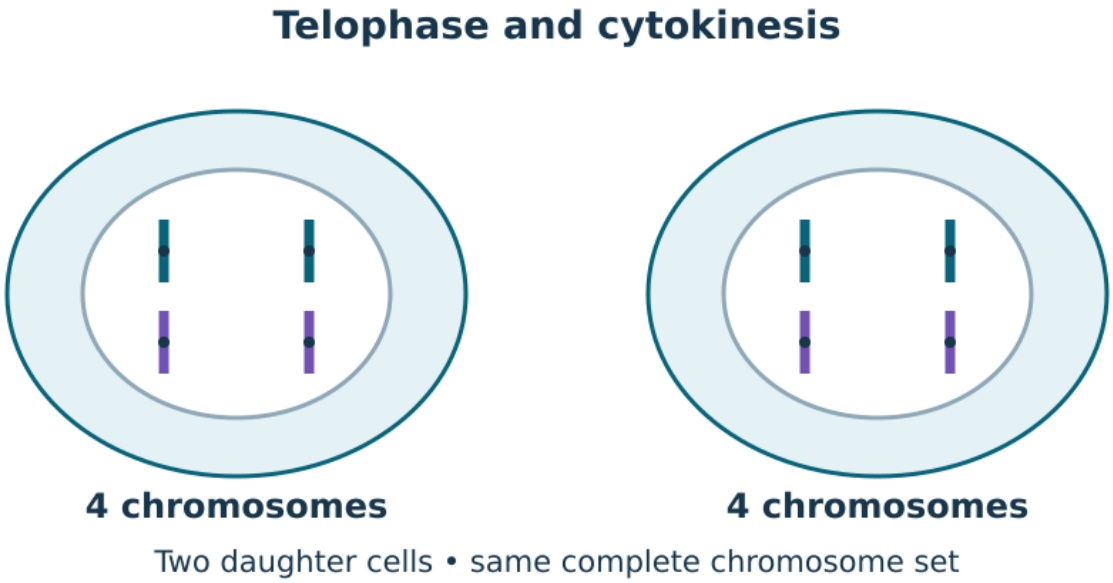
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Cells

Respond to the command word.

Select a value or mechanism that supports the answer.

Improve a claim by adding an appropriate limitation.



The DNA is copied before mitosis. The copies move apart into two sets. Each new cell gets a full set of genes.

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Evidence

Respond to the command word.

Select a value or mechanism that supports the answer.

Improve a claim by adding an appropriate limitation.

SIMULATED STUDY • invented teaching data

Size	Time	Unknown
20 people per group	8-week follow-up	Longer-term outcomes

A possible benefit is not a guarantee of safety or a cure.

Use numbers and a comparison before making a stem cell claim.

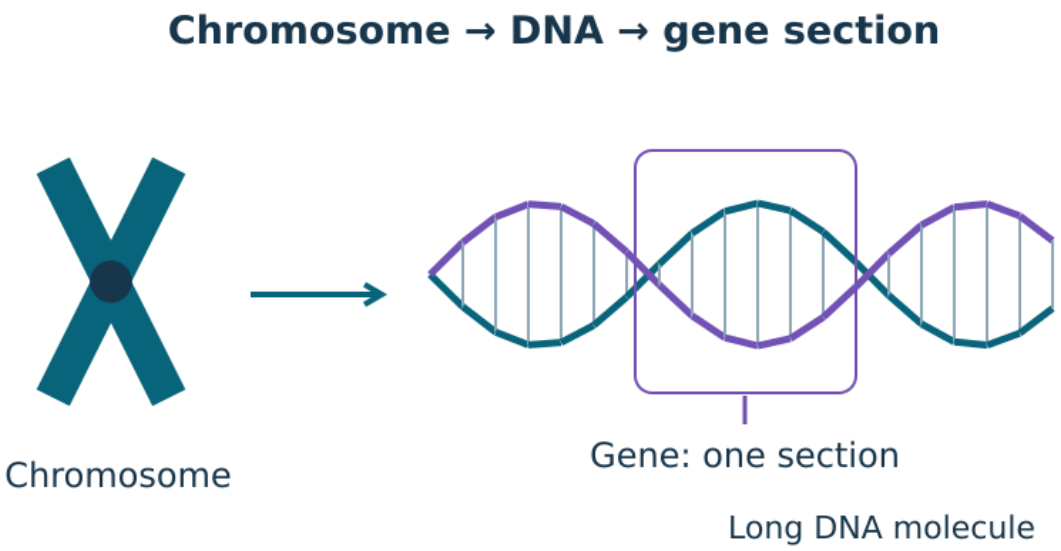
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Genetic information

Respond to the command word.

Select a value or mechanism that supports the answer.

Improve a claim by adding an appropriate limitation.



Connect nucleus, chromosome, DNA and gene; pair A–T and C–G.

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Inheritance

Respond to the command word.

Select a value or mechanism that supports the answer.

Improve a claim by adding an appropriate limitation.

Aa × Aa • one-gene model

Parent 1 gametes

	A	a
A	**AA**	**Aa**
a	**Aa**	**aa**

Parent 2 gametes

aa: 1/4 = 25% • chance for each fertilisation

Use gametes and a Punnett square, then interpret chance cautiously.

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We do · sort the cards

Respond to the command word.

Select a value or mechanism that supports the answer.

Improve a claim by adding an appropriate limitation.

Card	Your group
The length rose from 2.0 to 2.8 mm	_____
Chromosomes line up at the centre	_____
Copies separate so each nucleus receives a full set	_____
A root hair increases surface area for absorption	_____
The small sample limits generalisation	_____
A benefit is possible, but harms and follow-up matter	_____

Groups: Describe / Explain / Evaluate.

Choose first on your own. One partner explains and the other asks for the clue. Swap roles. Each learner keeps their own response.

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I do 2 · one example

Respond to the command word.

Select a value or mechanism that supports the answer.

Improve a claim by adding an appropriate limitation.

Find the percent for 8 out of 25.

8 ÷ 25 = 0.32.

0.32 × 100 = 32%.

In this sample, 32 in each 100 is the same share as 8 in 25.

The sample may differ from a wider group. We cannot give that group’s chance from this result alone.

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Three practice answers

Respond to the command word.

Select a value or mechanism that supports the answer.

Improve a claim by adding an appropriate limitation.

These are practice items, not the final test. Keep each first answer. Add a science link or a useful value, and qualify a claim that is too strong.

Prompt	Weak answer
Explain mitosis outcome	The cells match because they match
Calculate 7/20 as percent	It is about a third
Evaluate a short study	It proves the treatment is safe forever

My evidence and reasoning: _____

One limit or useful check: _____

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My improvement and next check

Respond to the command word.

Select a value or mechanism that supports the answer.

Improve a claim by adding an appropriate limitation.

First answer: _____

The value or model detail I used: _____

My science link: _____

My fair limit: _____

Keep your first answer. Hear the actual feedback. Make one real change, or give a reason to keep it. Record the next step you chose.

My next response: _____

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